Silicon & Handset — Living Ecosystem Engine, Emergent Differentiation & Autonomous Market Architecture

Document Role: Complete AI-Developer Implementation Blueprint (Physics, Component Vendor AI, Combinatorial BOM Architecture, Multi-Workload Benchmarking, Hedonic Pricing, and Stateful Consumer Simulation)
Target Codebase: `spinoff-phone` branch (`SPINOFF_SOURCE/`)
Owner Intent: The project owner is not a programmer or hardware engineer; they want a deeply realistic, watchable "living world" where every component maker and phone brand develops a unique identity, assembles genuinely different phones, competes on real physics and economics, prices autonomously, and sells to lifelike customers—100% driven by NPC decisions without player micromanagement.
Strict Guardrails (`AGENTS.md`):

- Never modify `AGENTS.md`, `SPINOFF_SOURCE/js/human-core.js` (SHA-256 `fa83eaa0ef3407f26f21675161f632a4365242f67fea7b0c192e59e75b9d5c32`), or `SPINOFF_SOURCE/CORE_LOCK.json`.
- Zero player operational commands (observer-only time controls, deep Thai inspection views, and logged `Edit Hub` condition experiments).
- Strict integer satang money conservation (`totalCash(w) === w.openingCapital + w.editCapital`), physical lot conservation, partition-invariant hourly ticks, read-only snapshot purity, and zero player-relative catch-up/rubber-banding.

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1. Core Architectural Philosophy: Why Current Phones Look Identical & How to Create True Emergence

In basic tycoon games (and in `v0.2.0`'s `kind:generation` model), technology is a 1D ladder (`Level 0 -> 1 -> 2 -> 3`), every supplier produces the same numbers, and buyers sort by a single `utility` scalar. That guarantees convergence: every firm builds the same phone.

In the real electronics industry, **differentiation emerges from four interacting forces**:

1. **Orthogonal Physical Trade-Offs (No Free Lunch):** Increasing CPU clock/cores raises peak wattage (P_{peak}) and die area (A_{die}, lowering wafer yield and raising cost). Increasing camera megapixels on a small sensor shrinks pixel pitch (μm), *worsening* low-light noise unless paired with a larger, thicker, costlier

optical module and strong ISP tuning. Packing a huge battery increases phone thickness and weight.

2. **Path-Dependent Corporate DNA:** Each supplier and handset brand has named founders and engineers (`HumanCore` traits + learned `subjects`) plus finite tooling. A chip firm staffed by low-power circuit specialists with a prudent CEO naturally evolves into an **efficiency/yield leader**, whereas an ambitious, high-risk chip firm chases **large-die flagship compute**.
3. **Combinatorial BOM + In-House Tuning:** A phone is not a "Level 2 Phone"; it is an explicit Bill of Materials combining 7–8 specific vendor SKUs (`SoC + DRAM + NAND + Camera + Display + Battery + Enclosure/Thermal + OS/BSP`) modified by the handset brand's own engineering tuning (PCB layout, thermal interface, ISP calibration, OS overhead optimization).
4. **Heterogeneous Customer Lives:** There is **no single "best phone."** A delivery rider working outdoors 10 hours/day values sunlight brightness, GPS/RF stability, battery endurance, and drop durability; a mobile gamer values 20-minute *sustained* (unthrottled) frame rates; a photography enthusiast values night dynamic range and storage speed/capacity.

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2. Recommended Engine Module Structure (`SPINOFF_SOURCE/`)

Keep compatibility with Node CommonJS tests (`npm test`) and standalone single-file bundling (`tools/build.cjs` -> `Silicon_Handset_single.html`) by organizing `SPINOFF_SOURCE/js/` into clean, deterministic UMD modules loaded in strict dependency order:

```
None
SPINOFF_SOURCE/js/
├── human-core.js            # IMMUTABLE LOCKED CORE (do not touch bytes/hash)
├── phone-engineering.js     # Existing baseline physics & role contracts
(preserve exports)
├── component-physics.js     # Parametric SKU synthesis, bus/mechanical
compatibility & 8-bench evaluator
├── supplier-ai.js           # Autonomous vendor R&D vectoring, lot production,
binning & dynamic quotes
├── brand-architect-ai.js    # Handset BOM solver, Make/Co-Tune/Buy logic &
Hedonic pricing engine
├── living-market.js         # Stateful consumer archetypes, pocketed installed
base, pain events & Logit choice
├── sim.js                   # Deterministic hourly/daily orchestrator, ledger,
serialize & strict audit()
└── ui.js                    # Read-only Thai Observer UI (Teardown Radar, BOM
Provenance, Market Pulse, Edit Hub)
```

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3. Component Vendor Simulation: Giving Chip, RAM, Storage, Camera & Display Firms Real Identity

3.1 Corporate DNA & Strategic Vector (`firm.dna`)

At world creation (and evolving slowly as leadership/staff change), derive each firm's strategic bias from its CEO and lead engineers' `HumanCore` profiles—never hardcoded scripts:

- `performanceBias`: weighted by `ambition + openness`
- `efficiencyBias`: weighted by `conscientiousness + prudence`
- `costYieldBias`: weighted by `prudence - riskTaking`
- `innovationRisk`: weighted by `riskTaking + openness`

3.2 Component-Specific Parametric Schemas (Distinct Engineering Physics per Category)

Every component design is an immutable `ComponentSKU` record `{ skuId, kind, designerFirmId, manufacturerFirmId, familyName, revision, unitDirectCostSatang, interfaceSpec, metrics, createdMinute }`:

Component Kind	**Physical & Interface Parameters (`metrics` & `interfaceSpec`)**	**Built-in Engineering Trade-Offs**
1. SoC / Chip (`chip`)	`isa`, `nodeNm`, `bigCores`, `littleCores`, `clockGhz`, `ipc`, `gpuGflops`, `npuTops`, `ispMaxMp`, `ispQuality`, `memBusType` (`SDRAM`/`LPDDR2`/`LPDDR3`/`LPDDR4`), `storageBus` (`SPI`/`eMMC`/`UFS`), `modemGen`, `dieAreaMm2`, `idleW`, `loadW`, `peakW`	Larger `dieAreaMm2` exponentially lowers Murphy wafer yield (`wafer()`); higher `clockGhz` scales `peakW` super-linearly ($P \propto C V^2 f$), punishing cheap enclosures with thermal throttling.
2. Volatile Memory (`ram`)	`memBusType`, `capacityMB`, `bandwidthGBps`, `latencyNs`, `idleRefreshMw`, `activeMw`	Must match `chip.memBusType` and `<= chip.maxRamMB`. Higher capacity increases

Component Kind	Physical & Interface Parameters (`metrics` & `interfaceSpec`)	Built-in Engineering Trade-Offs
		background self-refresh power; low bandwidth starves GPU/web scores even on a fast CPU.
3. Non-Volatile Storage (`storage`)	`storageBus` (`SPI`/`eMMC`/`UFS`), `nandType` (`SLC`/`MLC`/`TLC`/`QLC`), `capacityMB`, `seqReadMBps`, `seqWriteMBps`, `randIops`, `enduranceCycles`, `controllerEccScore`	Distinct from RAM! `QLC` offers cheap high `capacityMB` for photos/apps but lower `seqWriteMBps` and `enduranceCycles`; `UFS` + high `randIops` drives fast app launch and stutter-free camera burst.
4. Camera Module (`camera`)	`sensorAreaMm2`, `megapixels`, `pixelPitchUm` ($\sqrt{\text{area}/\text{MP}}$), `apertureF`, `lensElements`, `hasOis`, `afType` (`fixed`/`contrast`/`pdaf`), `zHeightMm`	**Physics realism:** Cranking `megapixels` without growing `sensorAreaMm2` shrinks `pixelPitchUm`, *reducing* low-light SNR! Large sensors + OIS require thick `zHeightMm` (camera bump / thicker chassis) and `chip.ispMaxMp >= megapixels`.
5. Display Panel (`display`)	`panelTech` (`TFT`/`IPS`/`OLED`/`LTPO`), `interface` (`RGB`/`MIPI`), `width`, `height`, `peakNits`, `contrastScore`, `refreshHz`, `minRefreshHz`, `wattsAt200Nits`, `wattsAtPeak`	High resolution + high `refreshHz` drains battery and taxes GPU fill-rate unless paired with costly `LTPO` variable refresh; cheap `TFT` fails outdoor sunlight legibility (`peakNits < 400`).
6. Battery & BMS (`battery`)	`chemistry` (`LiIon`/`LiPo`/`SiCarbon`), `capacityWh`, `volumeCm3`, `weightG`, `maxDischargeW`,	Must satisfy `maxDischargeW >= phonePeakPowerW * 1.05`. High `maxChargeW`

Component Kind	Physical & Interface Parameters (`metrics` & `interfaceSpec`)	Built-in Engineering Trade-Offs
	`maxChargeW`, `cycleLife80`, `bmsSafetyScore`	without high `bmsSafetyScore` and low chassis `rth` accelerates thermal aging and field RMA hazard.
7. Enclosure, Thermal & RF (`enclosure`)	`material` (`plastic`/`aluminium`/`glass_al`/`titanium`), `coolingTech` (`none`/`graphite`/`heatpipe`/`vapor_chamber`), `internalVolumeCm3`, `maxModuleZMm`, `rthCW`, `thermalMassJC`, `weightG`, `thicknessMm`, `dropScore`, `ipCode`, `rfAttenuationDb`	Metal chassis lowers `rthCW` (cooler skin/less throttling) and raises `dropScore`, but increases `rfAttenuationDb` unless paired with precision antenna windowing (`pcb_rf` quality)! Thin chassis restricts `battery.volumeCm3` and `camera.zHeightMm`.

3.3 Autonomous Supplier Competition Loop

Suppliers are independent profit-seeking businesses (`supplier-ai.js`):

1. **Market Signal Observation:** Each week, every supplier inspects RFQ win/loss logs (`lostDueToPrice`, `lostDueToSpec`, `lostDueToStockout`, `lostDueToBusIncompatibility`).
2. **Endogenous R&D Direction:**
    - If losing orders on price to a direct rival -> start a **Cost/Yield Reduction Revision** (die shrink, layout optimization, or value-tier SKU).
    - If handset brands are bottlenecked on heat/battery -> a prudent supplier researches **Low-Leakage / High-Efficiency SKUs**.
    - If high-margin flagship cohorts have unmet demand -> an ambitious supplier with strong `semiconductor`/`optics` talent researches a **Next-Gen Interface / High-Spec SKU**.
3. **Dynamic Bid/Ask Quote Engine & Obsolescence Pruning:**
    - Replace static `priceBps` with dynamic inventory-cover pricing: suppliers trim margins (`-3%` to `-6%`) on slow-moving lots to clear working capital, raise margins (`+4%` to `+10%`) when capacity utilization exceeds `85%`, and **halt manufacturing of obsolete SKUs** once downstream brand orders shift to newer buses.

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4. How Handset Brands Assemble Distinct Phones (`brand-architect-ai.js`)

When a handset brand initiates a new phone project, it executes a **4-Step Autonomous Product Architecture Pipeline**:

Step 1: Strategic Positioning Selection

The executive team evaluates all consumer archetypes' current unmet pain points, rival active phones, and its own cash runway + engineering strengths (`optics`, `software`, `semiconductor`, `electronics`, `manufacturing`), selecting a **Product Thesis** (e.g., *Endurance Value Workhorse*, *Low-Light Camera Specialist*, *Affordable Esports Performer*, *Balanced Mainstream Flagship*).

Step 2: Hard Engineering & Mechanical Compatibility Filtering

Candidate supplier SKUs (from qualified open-market quotes + any proprietary in-house SKUs owned by the brand) are filtered by real physical compatibility:

- Bus match: `ram.memBusType === chip.memBusType`, `storage.storageBus === chip.storageBus`, `display.interface === chip.displayInterface`.
- Power & ISP envelope: `battery.maxDischargeW >= estimatedPeakW * 1.05`, `camera.megapixels <= chip.ispMaxMp`.
- Mechanical package fit: `battery.volumeCm3 + boardVolumeCm3 <= enclosure.internalVolumeCm3` and `camera.zHeightMm <= enclosure.maxModuleZMm`.

Step 3: Bounded Combinatorial BOM Optimization

Instead of greedy single-part picking, evaluate top candidate BOM combinations `(chip, ram, storage, camera, display, battery, enclosure)` within the target BOM budget ceiling: $$\text{BOMScore}(\mathbf{b}) = \hat{U}*{\text{target}}(\mathbf{b}, \text{firmTuning}) - \lambda*{\text{cost}} \cdot \text{TotalQuoteCost}(\mathbf{b}) - \lambda_{\text{delay}} \cdot \max_i(\text{LeadTime}*i) - \lambda*{\text{risk}} \cdot \text{SingleSourceRisk}(\mathbf{b})$$

- **Why Two Brands Buying the Same Chip Still Build Different Phones:**
  1. They pair that chip with different RAM/NAND/Camera/Cooling/Battery SKUs based on their Product Thesis and CEO traits.
  2. **In-House Engineering Tuning (`firmTuning`):** During `research` and `prototype` stages, the brand's own engineers imprint measurable proprietary gains:

- `optics` + `software` staff calibrate ISP tone-mapping/denoise (`+ispTuningFactor`).
- `software` staff optimize kernel/BSP (`-osOverhead`, `-idleWakeupMw`).
- `electronics` + `manufacturing` staff optimize PCB layout and thermal contact (`-effectiveRth`, `+rfEfficiency`).

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5. Measuring "Which Phone Is Better" Without Fake Universal Scores (`component-physics.js`)

Never collapse a phone into a single magic number! Compute an **8-Axis Objective Engineering Benchmark Vector** under standardized physical workloads:

1. **Connectivity & Call Stability (`bench.connectivity`, 0–100):** Function of `chip.modemGen`, `pcb_rf.signalIntegrity`, minus uncompensated `enclosure.rfAttenuationDb`.
2. **Everyday Responsiveness & Multitasking (`bench.dailyFluidity`, 0–100):** $$\text{Fluidity} = f\Big(\text{SingleCorePerf},; \text{ram.bandwidthGBps},; \text{storage.randIops},; \text{clamp}\big(\frac{\text{ram.capacityMB} - \text{osFootprintMB}}{\text{appWorkingSetMB}}\big),; 1 - \text{osOverhead}\Big)$$ *(Captures real life: a phone with a fast CPU but slow eMMC storage or cramped RAM stutters in daily apps!)*
3. **Peak 3D/Compute vs. 20-Minute Sustained Performance (`bench.peakPerf`, `bench.sustainedPerf`):**
   - Steady-state skin temp under gaming load: $T_{\text{skin}} = T_{\text{ambient}} + P_{\text{game}} \cdot R_{\text{th,eff}}$.
   - If $T_{\text{skin}} > 43^\circ\text{C}$, DVFS thermal throttling scales sustained clocks by $\theta_{\text{throttle}} = \text{clamp}\big((\frac{43 - T_{\text{ambient}}}{T_{\text{skin}} - T_{\text{ambient}}})^{0.85},; 0.35,; 1.0\big)$, moderated over short bursts by `enclosure.thermalMassJC`.
   - `sustainedPerf = peakPerf * throttleRatio` — exposing "spec-sheet monsters" that overheat in 3 minutes!
4. **Daylight Detail & Night/Portrait Photography (`bench.photoDay`, `bench.photoNight`, 0–100):**
   - `photoDay` rewards `megapixels`, `lensElements`, `afType`, and `ispQuality`.
   - `photoNight` rewards **photon gathering** ($\text{pixelPitchUm}^2 / \text{apertureF}^2$), `hasOis` (`+22%` shutter-time advantage), `chip.npuTops` / `ispQuality`, and brand `ispTuningFactor`.
5. **Mixed-Use Battery Life & Fast-Charge Speed (`bench.batteryHours`, `bench.chargeMinutes`):**

- `batteryHours` = `0.92 * battery.capacityWh / mixedDutyWatts` (combining standby leakage, screen nits power, cellular modem drain, and CPU/RAM active duty).
- `chargeMinutes` = `60 * (0.80 * battery.capacityWh) / effectiveChargeWatts` (derated if thermal resistance causes charge throttling).

6. **Display Visual Quality & Sunlight Readability (`bench.displayVisual`, `bench.sunlightReadability`):** Driven by PPI sharpness, `contrastScore` (`OLED` > `IPS` > `TFT`), `refreshHz`, and `peakNits` (`>= 800` nits needed for strong outdoor readability).
7. **Ergonomics & Build Craft (`bench.ergonomics`, 0–100):** Rewards slimness (`7.2–8.5mm`), balanced weight (`145–190g`), cool operating skin temp, and tactile chassis material; penalizes heavy/thick bricks (`> 225g` or `> 10.5mm`).
8. **Durability, Reliability & Longevity (`bench.reliability`, `expectedLifespanDays`):** Driven by `enclosure.dropScore`, `ipCode`, `battery.cycleLife80`, `storage.enduranceCycles`, `bmsSafetyScore`, and `os.supportMonths`—directly governing post-sale field failure hazard (`RMA`) and second-year customer satisfaction!

Also output a plain-Thai **`engineeringDiagnosisTh`** string derived strictly from the lowest bottleneck terms (e.g., *"ชิปประมวลผลแรงแต่ตัวถังพลาสติกระบายความร้อนไม่ทัน ทำให้เฟรมเรตตก 32% เมื่อเล่นต่อเนื่อง"* or *"เซนเซอร์รับแสงพิกเซลใหญ่พร้อมกันสั่น OIS ถ่ายภาพกลางคืนโดดเด่น แต่กินพื้นที่ตัวเครื่องหนา 9.4 มม."*).

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6. How NPCs Determine Price Autonomously (Hedonic Equilibrium & Dynamic Margin Engine)

Eliminate arbitrary static markups and rigid sunk-cost floors. Price is set and updated weekly by each brand's autonomous commercial logic in **four transparent layers**:

1. **Full Unit Economic Cost Calculation (`fullUnitCostSatang`):** $$\text{UnitCost} = \underbrace{\sum_{i \in \text{BOM}} \text{ActualLotUnitCost}*i* + *\text{EMSAssemblyFee}* + *\text{RoyaltySatang}*}{\text{Direct Marginal Cost } (C_{\text{marginal}})} + \underbrace{\frac{\text{ProjectNRE}}{\text{TargetAmortizationUnits}} + \text{ExpectedWarrantyRMASatang}}_{\text{Amortized & Lifecycle Provision}}$$
2. **Hedonic Market Benchmark Price (P_{hedonic}):** Compare the phone's target-cohort utility $U_{\text{self}, s}$ against all currently in-stock rival phones $k \in \mathcal{M}_s$:

- Compute market value-per-utility ratio $\rho_s = \text{median}*k \left(\frac{P_k}{U}{k,s}\right)$ (falling back to cohort willingness-to-pay anchor if the market is empty).
- $P_{\text{hedonic}} = \rho_s \cdot U_{\text{self}, s} \cdot \left(0.88 + 0.24 \cdot \frac{\text{BrandTrust}}{100}\right)$.

3. **Launch Price Positioning:**
   - **Premium / Skimming Strategy** (high CEO `ambition` + unique spec lead): price near $\max(\text{UnitCost} \cdot 1.25,; P_{\text{hedonic}} \cdot 1.08)$.
   - **Value / Penetration Strategy** (high market-share focus or challenger brand Trust `< 50`): price at $0.90 \cdot P_{\text{hedonic}}$ (as long as $\ge C_{\text{marginal}} \cdot 1.06$).
4. **Weekly Dynamic Repricing (Inventory Velocity & Runway Feedback):**
   - **Scarcity / Hot Seller (`daysOfInventory < 5` and high conversion):** Hold or nudge price up `+3%` to capture margin.
   - **Slow Mover (`daysOfInventory > 25`):** Trim price `-5%` toward $C_{\text{marginal}} \cdot 1.03$.
   - **Obsolete Model or Liquidity Distress (`isSuperseded` or `cashRunwayDays < 14`): Drop the sunk-cost floor** and allow clearance markdowns down to `0.55 * C_marginal` to recover urgent cash!

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7. Making Customers Feel Truly Alive (`living-market.js`)

Achieve rich human behavior at $O(1)$ per-cohort speed (zero browser lag) using **8 Stateful Consumer Archetypes**:

Archetype ID	Thai Persona Name	Core Weight Profile (What They Care About)	Typical Upgrade Trigger
`basic_durable`	ผู้ใช้ทั่วไปเน้นทนทานและประหยัด	Price (35%), Battery (25%), Connectivity (20%), Reliability (20%)	Old phone broken or battery dies
`outdoor_worker`	คนทำงานภาคสนามและไรเดอร์	Battery (30%), Sunlight Display (25%), Connectivity/GPS (25%), Drop/IP (20%)	Dim screen, weak signal, or battery degradation

Archetype ID	Thai Persona Name	Core Weight Profile (What They Care About)	Typical Upgrade Trigger
`social_photo`	สายโซเชียลและถ่ายภาพคาเฟ่/กลางคืน	Night/Day Camera (45%), Storage Capacity (20%), OLED Display (20%), Ergonomics (15%)	Camera leap ($\Delta \text{Photo} > 18$) or storage full
`mobile_gamer`	เกมเมอร์มือถือสายแข่ง	**Sustained** Perf (45%), Display Hz/Touch (20%), Battery/Charge (20%), Price Value (15%)	Thermal throttling / FPS drop on new app demands
`pro_exec`	นักธุรกิจและคนทำงานมืออาชีพ	Daily Fluidity (30%), Reliability/Security (25%), Brand Trust (25%), Thin/Light Craft (20%)	OS security end-of-life or flagship cycle
`student_value`	นักเรียนนักศึกษาสายคุ้มสเปก	Price-to-Fluidity (40%), Peak/Sustained Perf (25%), Camera (20%), Storage (15%)	High value-per-baht challenger launch
`media_binger`	สายดูหนังซีรีส์และสตรีมมิง	OLED Contrast/Size (35%), Battery Hours (35%), Storage (15%), Fluidity (15%)	Screen/battery upgrade
`tech_enthusiast`	สายไอทีชอบลองนวัตกรรมใหม่	Frontier Silicon/Camera/OS Innovation (55%), Sustained Perf (25%), Display (20%)	Early adopter of novel custom silicon/optics

How Customer Life & Choice Work Each Day

1. **Pocketed Device Experience:** Each archetype tracks vintage buckets of phones they currently own (`installedBase`). Every day, aging reduces battery capacity and

accumulates storage wear; hot/fragile phones trigger **Field Failures (RMA)** and concrete **Pain Logs** (e.g., *"กลุ่มเกมเมอร์บ่นรุ่น X เครื่องร้อน 47°C จนเฟรมเรตตกหลังเล่น 10 นาที"*).

2. **Word-of-Mouth Reputation (`brandTrust`):** When a phone delivers higher real experience than its price class and low RMA rates, owners generate positive word-of-mouth (`brandTrust` rises slowly). High RMA failures or severe thermal throttling slash `brandTrust`—directly hurting that brand's next launch!
3. **Endogenous Upgrade + Multinomial Logit Choice:**
    - Buyers enter the market when first-time adoption or aging/pain on their current pocketed phone exceeds threshold, **funded strictly by conservation-exact circular household income** (recycled from wages + upstream factor payments).
    - Each funded buyer evaluates affordable, in-stock phones using **Net Surplus Utility**: $$V_{j, a} = \mathbf{w}*a \cdot \mathbf{Bench}(j)* + \gamma{\text{trust}, a} \cdot \text{BrandTrust}(\text{firm}*j)* - \alpha{\text{price}, a} \cdot \ln(P_j) - U_{\text{incumbent}, a}$$
    - Selection uses **Seeded Multinomial Logit** across all viable phones plus an **Outside Option ("Wait & Keep Cash" $V_0 = 0$)**—so overpriced or stagnant generations naturally see buyers hold onto their old phones!

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8. Observer UI Additions (Making Differences Visible at a Glance in Thai)

Update `SPINOFF_SOURCE/js/ui.js` (preserving mobile `390px` and desktop `1280px` cleanliness and read-only purity):

1. **Component Ecosystem Matrix (แท็บซัพพลายเชนและตลาดชิ้นส่วน):** Show every Chip, RAM, Storage, Camera, Display, Battery, and Enclosure company's Corporate DNA badge, active SKUs, specs, current dynamic quote price, stock, and which handset brands buy from them.
2. **BOM Provenance & Trade-Off Card (การ์ดผ่าไส้ในมือถือแต่ละรุ่น):** For every phone, display the exact vendor + SKU of all 7 components, why the brand's engineers chose that combination, its **8-Axis Benchmark Profile**, its **Thermal Throttling Ratio**, and its **Pricing Breakdown** (`BOM + Assembly + NRE + Margin` vs. `Market Hedonic Price`).
3. **Customer Voice & Installed-Base Pulse (เสียงผู้ใช้งานจริงและเครื่องในตลาด):** Show which consumer personas bought which models, what phones are currently in people's pockets, real pain-point quotes generated from physics bottlenecks, RMA rates, and `brandTrust` trajectories.

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9. Recommended Implementation Sequence for the Coding AI

To ship cleanly without breaking `audit()` or existing regression suites, implement in **three tightly tested commits/stages** (or a single cohesive major release `v0.4.0` if `v0.3.0` is already merged):

1. **Stage A — Parametric Component Physics & Combinatorial BOM Engine (`component-physics.js` + `brand-architect-ai.js`):** Implement the 7-category parametric schemas, hard compatibility solver, firm tuning modifiers, and 8-axis physical benchmarks. Verify with unit tests that identical chips in different enclosures/RAM/cameras yield measurably distinct benchmark vectors.
2. **Stage B — Autonomous Supplier AI & Hedonic Dynamic Pricing (`supplier-ai.js` + pricing loop):** Wire supplier DNA, RFQ win/loss R&D steering, dynamic quote pricing, obsolescence halting, and 4-layer brand hedonic/clearance pricing.
3. **Stage C — Living Consumer Archetypes, Circular Cash Flow & Thai Teardown UI (`living-market.js` + `sim.js` + `ui.js`):** Replace winner-take-all `candidates[0]` with 8 stateful archetypes, pocketed `installedBase` pain/RMA feedback, `brandTrust`, seeded Multinomial Logit + Outside Option, circular satang income recycling, and updated Thai observer cards. Run `npm run check`, `npm test`, `npm run build`, `npm run test:browser`, and `node tools/soak.cjs` (`5 seeds × 1,095 days`).